

Lirui Wang

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EDUCATION

University of Washington, Seattle, WA

Bachelor of Science, Electrical Engineering (Honors), College of Engineering

Bachelor of Science, Computer Science (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics (Minor), College of Arts and Sciences

Master of Science, Computer Science, College of Arts and Sciences

Sep. 2016 – Present.

GPA: 3.96 / 4.00 (*summa cum laude*), Advisor: Prof. Dieter Fox

RESEARCH PROJECTS

Deep Visuomotor Policies for Cluttered Scene Grasping, Seattle, WA

UW Robotics and States Estimation Lab

Jan. 2020 – Present.

- Introduced goal auxiliary actor-critic methods for 6D grasping from point clouds in cluttered scenes.

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection, Seattle, WA

UW Robotics and States Estimation Lab

March. 2019 – Dec. 2019

- Introduced an integrate planner that optimizes manipulation trajectory with online grasp synthesis and goal selection.

Machine Learning in Robot Trajectory Optimization, Seattle, WA

Statistical Learning Course Project

Sep. 2019 – Dec. 2019

- Studied trajectory optimization as functional gradient descent using RKHS and as probabilistic inference using GP.

6D Object Pose Tracking with the Deep Iterative Matching Network, Seattle, WA

UW Robotics and States Estimation Lab

June. 2018 – March. 2019

- Extended DeepIM to robustly track robot arm and YCB objects for occluded scenes and AR applications.

Unsupervised Flow and Depth Methods Study, Seattle, WA

UW Robotics and States Estimation Lab

Jan. 2018 – Jun. 2018

- Investigated unsupervised learning methods for 3D geometry tasks such as depths and flow on a robotic setting
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EXPERIENCE

Applied Research Intern, NVIDIA, WA

June. 2020 – Oct. 2020

- Researching predictive model for model-based RL in autonomous driving scenarios

Industry Capstone Project, Microsoft, WA

Jan. 2020 – June. 2020

- Built on autonomous mobile manipulator to pull out servers from server rack and transport it to repair.

Teaching Assistant, University of Washington, WA

Sep. 2019 – Mar. 2020

- Teach CSE 455 Computer Vision, CSE 473 Artificial Intelligence, and CSE 490/599 G1 Deep Learning.

Research Assistant, UW Robotics and State Estimation Lab, WA

Sep. 2018 – Present.

- Worked on manipulation research projects with Dr. Yu Xiang and Prof. Dieter Fox

Robotics Engineer Intern, DJI Technology, China

Jun. 2017 – Aug. 2017

- Worked on an automatic robot battle that composes of a vehicle navigated by SLAM and a DJI Manifold with Guidance
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PUBLICATIONS

Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds, Lirui Wang, Yu Xiang, Dieter Fox

Under Review

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection, Lirui Wang, Yu Xiang, Dieter Fox

In Proceedings of Robotics: Science and Systems (RSS 2020)

SKILLS & OTHERS

Programming Language: Java, C++/C, Python, Matlab

Tools: ROS, Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt, SLAM

Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing, Statistical Learning, High-Dimension Statistics, Robust Machine Learning, Reinforcement Learning, Interactive Learning

Scholarships: Tau Beta Pi Scholarship 2017-2019, Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship 2017-2018, Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship 2018-2019